

Chapter 6: Pressure, Winds, Storms, and Cyclones

— Complete Chapter Solutions

NCERT Class 8 Science (Curiosity)

Q1. Why are winds stronger on some days than on others?

Answer: Wind is created by the movement of air from a region of high air pressure to a region of low air pressure. The greater the difference in air pressure between two locations, the faster and stronger the wind blows. On days with large pressure differences (e.g., during approaching weather fronts or storms), winds become significantly stronger.

Q2. Why are water tanks usually placed at a height?

Answer: Liquid pressure increases with the height (depth) of the liquid column above the point of exit. By placing a water tank on the roof or at a high elevation, a tall column of water is created in the supply pipes. This high gravitational pressure forces water down through the pipes and produces a strong, steady stream at the taps below without requiring a pump.

Q3. Can air pressure really crush us?

Answer: No. Although atmospheric pressure exerts a huge force on our bodies (about 101.3 k Pa or $1.01 \times 10^5 \text{ N/m}^2$ at sea level), the fluids and gases inside our body cells exert an equal and opposite outward pressure. This internal pressure balances the external atmospheric pressure, preventing us from being crushed.

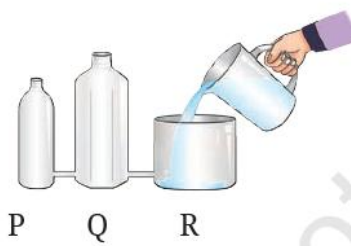
Q4. What causes storms and cyclones? If the Earth stopped rotating, would cyclones still form?

Answer: Cause: Over warm ocean waters, intense heating causes warm, moist air to rise rapidly, creating a deep low-pressure region. As surrounding air rushes in and rises, condensation of water vapor releases latent heat, further warming the air and intensifying the low-pressure zone.

If Earth stopped rotating: Cyclones would NOT form. The spinning motion of cyclones is caused by the Earth's rotation (Coriolis Effect). Without Earth's rotation, air would move straight into the low-pressure zone without spinning, so standard storms could occur, but rotating cyclones could not form.

Q1. Choose the correct option:

(i) Look at Fig. 6.21 carefully. Vessel R is filled with water. When pouring of water is stopped, the level of water will be:



- * (a) the highest in vessel P
- * (b) the highest in vessel Q
- * (c) the highest in vessel R
- * (d) equal in all three vessels

Correct Answer: (d) equal in all three vessels

Reasoning: Water flows between interconnected vessels until the pressure at the bottom of each column is balanced. Liquid pressure depends solely on the height of the liquid column, not the shape or width of the container.

(ii) A rubber sucker (M) is pressed on a flat smooth surface and an identical sucker (N) is pressed on a rough surface:

- (a) Both M and N will stick to their surfaces.
- (b) Both M and N will not stick to their surfaces.
- (c) M will stick but N will not stick.
- (d) M will not stick but N will stick.

Correct Answer: (c) M will stick but N will not stick.

Reasoning: Sucker M creates an airtight seal on the smooth surface, squeezing out air inside and allowing atmospheric pressure from the outside to hold it tightly. On the rough surface, microscopic gaps allow outside air to leak under sucker N, eliminating the pressure difference.

(iii) A water tank is placed on the roof of a building at a height 'H'. To get water with more pressure on the ground floor, one has to:

- (a) increase the height 'H' at which the tank is placed.
- (b) decrease the height 'H' at which the tank is placed.
- (c) replace the tank with another tank of the same height that can hold more water.
- (d) replace the tank with another tank of the same height that can hold less water.

Correct Answer: (a) increase the height 'H' at which the tank is placed.

Reasoning: Liquid pressure depends directly on the height of the water column ($P = h \rho g$), independent of the total volume or capacity of the tank.

(iv) Two vessels, A and B, contain water up to the same level as shown in Fig. 6.22. P_A and P_B represent the pressure at the bottom, and F_A and F_B represent the force exerted by water at the bottom. Base area of B is larger than base area of A:

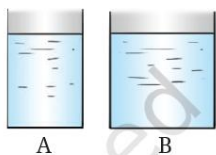


Fig. 6.22

- (a) $P_A = P_B, F_A = F_B$
- (b) $P_A = P_B, F_A < F_B$
- (c) $P_A = P_B, F_A = F_B$
- (d) $P_A = P_B, F_A = F_B$
- (e) $P_A = P_B, F_A > F_B$

Correct Answer: (b) $P_A = P_B, F_A < F_B$

Reasoning: Since both vessels hold water up to the exact same depth/level (h), the liquid pressure at the bottom is identical ($P_A = P_B$). However, Force = Pressure $\times$ Area ($F = P.A$). Since vessel B has a larger base area ($A_B > A_A$), the total downward force exerted on the base of B is greater ($F_A < F_B$).

Q2. State whether the following statements are True [T] or False [F]:

(i) Air flows from a region of higher pressure to a region of lower pressure.

Answer: True [T]

(ii) Liquids exert pressure only at the bottom of a container.

Answer: False [F] (Liquids exert pressure in all directions, including against the side walls of containers).*

(iii) Weather is stormy at the eye of a cyclone.

Answer: False [F] (The "eye" at the center of a cyclone is a calm, low-pressure region with light winds and clear skies; the violent storm is in the surrounding eyewall).

(iv) During a thunderstorm, it is safer to be in a car.

Answer: True [T] (The metal frame of a car acts as a Faraday cage, safely conducting any lightning strike around the exterior to the ground).

Q3. Fig. 6.23a shows a boy lying horizontally, and Fig. 6.23b shows the boy standing vertically on a loose sand bed. In which case does the boy sink more in sand? Give reasons.

- **Answer:** The boy sinks much more when **standing vertically** on the loose sand bed.
- **Reasoning:**

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$

The weight (force) of the boy is the same in both cases. When standing vertically, this force acts over a very small surface area (only the soles of his feet). This smaller contact area results in **higher pressure** exerted on the sand, causing his feet to sink deeper. When lying down, his weight is distributed over a much larger surface area, resulting in **lower pressure**.

Q4. An elephant stands on four feet. If the area covered by one foot is 0.25m^2 , calculate the pressure exerted by the elephant on the ground if its weight is $20,000\text{N}$.

- **Given:**
 - Weight of elephant (Force, F) = $20,000\text{N}$
 - Area of 1 foot = 0.25m^2
 - Total area of 4 feet (A) = $4 \times 0.25\text{m}^2 = 1.0\text{m}^2$
- **Formula:**
- Pressure (P) = $\frac{\text{Force}(F)}{\text{Area}(A)}$

- **Calculation:**

$$P = \frac{20,000\text{N}}{1.0\text{m}^2} = 20,000\text{ Pa (or N/m}^2\text{)}$$

- **Answer:** The pressure exerted by the elephant on the ground is **20,000 Pascal**.

Q5. There are two boats, A and B. Boat A has a base area of 7m^2 , and 5 persons are seated in it. Boat B has a base area of 3.5m^2 , and 3 persons are seated in it. If each person has a weight of 700N , find out which boat will experience more pressure on its base and by how much?

- **Calculation for Boat A:**
 - Total weight (Force, F_A) = $5 \times 700\text{N} = 3500\text{N}$
 - Base Area (A_A) = 7m^2
 - Pressure (P_A) = $\frac{3500\text{N}}{7\text{m}^2} = 500\text{ Pa}$
- **Calculation for Boat B:**
 - Total weight (Force, F_B) = $3 \times 700\text{N} = 2100\text{N}$
 - Base Area (A_B) = 3.5m^2
 - Pressure (P_B) = $\frac{2100\text{N}}{3.5\text{m}^2} = 600\text{ Pa}$
- **Comparison:**

$$\text{Difference} = P_B - P_A = 600\text{ Pa} - 500\text{ Pa} = 100\text{ Pa}$$

- **Answer: Boat B** experiences more pressure on its base than Boat A by **100 Pa**.

Q6. Would lightning occur if air and clouds were good conductors of electricity? Give reasons for your answer.

- **Answer: No, lightning would not occur.**
- **Reasoning:** Lightning requires the buildup of massive static electric charges (positive at the top of clouds, negative at the bottom) over time. Normally, air acts as an **electrical insulator**, keeping these opposite charges separated until a critical voltage breakdown triggers a massive sudden discharge. If air and clouds were good conductors, electrical charges would continuously flow and neutralize each other smoothly as soon as they formed, preventing any charge accumulation or sudden lightning discharge.

Q7. What will happen to the two identical balloons A and B (placed at the same depth in a plastic bottle) when water is filled into the bottle up to a certain height? Will both balloons bulge? If yes, will they bulge equally? Explain your answer.

- **Answer: Yes, both balloons will bulge, and they will bulge equally.**
- **Reasoning:** Liquids exert pressure equally in all directions at the same horizontal level/depth. Since balloons A and B are situated at identical depths below the water surface, the water pressure acting on both balloon membranes is identical, causing them to bulge outward to the same extent.

Q8. Explain how a storm becomes a cyclone.

- **Answer:**
 1. **Warm Ocean Heating:** Over warm ocean waters (above 26.5°C), the air gets heated and absorbs large amounts of moisture.
 2. **Rising Air & Low Pressure:** This warm, moist air expands, grows lighter, and rises rapidly, creating a region of very low pressure near the sea surface.
 3. **Inrushing Air:** Cooler air from neighboring high-pressure areas rushes in toward the center to replace the rising air.
 4. **Latent Heat Release:** As the rising air cools at higher altitudes, water vapor condenses into rain droplets, releasing latent heat into the atmosphere. This heat warms the surrounding air, forcing it to rise even higher and lowering the central pressure further.
 5. **Swirling Motion (Coriolis Effect):** Due to the Earth's rotation, the intruding winds begin to spin into a spiral.

6. **Cyclone Formation:** This continuous cycle develops into a large, spinning storm system with extremely high wind speeds rotating around a calm, ultra-low-pressure center called the **eye**.

Q9. Fig. 6.25 shows trees along the sea coast in a summer afternoon. Identify which side is land, A or B. Explain your answer.

- **Answer: Side B is the land, and Side A is the sea.**
- **Explanation:**
 - During a sunny summer afternoon, land heats up much faster than water.
 - The air above the land (Side B) becomes warm, expands, and rises, creating a low-pressure area.
 - Cooler, higher-pressure air over the sea (Side A) rushes toward the land to replace it, creating a **sea breeze** that blows from **A to B**.
 - Since the trees are bending/leaning from Side A toward Side B, the wind is blowing from A to B, confirming that **B is the land side**.

Q10. Describe an activity to show that air flows from a region of high pressure to a region of low pressure.

- **Materials Required:** Two identical rubber balloons, a hollow drinking straw, and thread.
- **Procedure:**
 1. Inflate one balloon fully so that it contains high-pressure air inside. Tie its mouth to one end of the drinking straw using thread.
 2. Leave the second balloon uninflated (or partially inflated) so that the air pressure inside it is significantly lower.
 3. Attach the other end of the straw to the neck of the second balloon.
 4. Release the pinch/clamp on the straw connecting the two balloons.
- **Observation:** Air immediately flows from the fully inflated balloon into the uninflated balloon until the air pressures inside both balloons become equal.
- **Conclusion:** This demonstrates that air naturally flows from a region of higher pressure to a region of lower pressure.

Q11. What is a thunderstorm? Explain the process of its formation.

- **Definition:** A **thunderstorm** is a weather event characterized by heavy rain, strong gusts of wind, lightning, and thunder.
- **Formation Process:**
 1. In hot and humid conditions, warm, moist air rises rapidly (updrafts).

2. As the air reaches high altitudes, it cools, causing water vapor to condense into clouds and heavy water droplets/ice crystals.
3. The water droplets fall downward while strong updrafts continue pushing moist air upward.
4. The violent friction between upward-moving ice crystals and downward-moving water droplets separates electric charges in the cloud.
5. When charge separation becomes massive, electric discharges occur as **lightning**, creating intense heat that rapidly expands the air, generating the sound wave known as **thunder**.

Q12. Explain the process that causes lightning.

- **Answer:**
 1. **Charge Separation:** Inside storm clouds, violent collisions between rising ice crystals and falling water droplets cause electric charges to separate. Lighter, positively charged ice crystals accumulate at the upper cloud edge, while heavier, negatively charged water droplets collect near the bottom of the cloud.
 2. **Ground Induction:** The large negative charge at the cloud base induces a positive charge on the ground below (trees, buildings, open fields).
 3. **Insulation Breakdown:** Air normally acts as an insulator preventing charge flow. However, when the electric charge difference becomes extremely large, the insulating property of air breaks down.
 4. **Sudden Discharge:** A massive stream of negative charge surges toward the positive ground (or between clouds) in a fraction of a second. This intense electric current heats the air path up to $30,000^{\circ}\text{C}$, producing a brilliant flash of light (**lightning**).

Q13. Explain why holes are made in banners and hoardings.

- **Answer:** Holes are made in roadside banners and hoardings to allow high-speed wind to pass through them easily.
- **Reasoning:** When strong winds blow, a solid banner traps air, creating a large pressure difference between its front and back surfaces. This large force can tear the fabric or collapse the supporting frame. The holes reduce air resistance and pressure buildup on the surface, allowing the banner to remain intact during windy weather.